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November 12, 1973
SUPERSEDING
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August 16, 1966

FEDERAL SPECIFICATION

HEELS, RUBBER

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers heels made of natural rubber, synthetic rubber, or a mixture of these materials.

1.2 Classification.

1.2.1 Types, classes, and styles. The rubber heels shall be of the following types, classes, and styles, as specified (see 6.2).

Type I - Half heels.

Class:
- 1 - Standard composition.
- 3 - Nonmarking composition.
- 5 - Deleted.
- 6 - Deleted.
- 7 - Deleted.
- 8 - Deleted.

Style 1 - Straight breast.

Type II - Whole heels.

Class:
- 1 - Standard composition.
- 2 - Oil resistant composition.
- 3 - Nonmarking composition.
- 4 - Oil resistant, nonmarking, cork composition.
- 5 - Wood core heel, oil resistant, nonmarking composition.
- 6 - Deleted.
- 7 - Plastic core heel, oil resistant, nonmarking composition.
- 8 - Deleted.

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Style:

1 - Straight breast.
2 - Full beveled breast.
3 - Straight breast truncated corners.

Type III - Junior and cuban heels.

Class:
1 - Deleted.
3 - Nonmarking composition.

Style 1 - Straight breast.

1.2.1.1 Subclass (a). When specified (see 6.2), type I, class 3 heels shall be supplied in high abrasive composition subclass (a).

1.2.2 Sizes.

1.2.2.1 Types I and II. Heels furnished to shoe manufacturers for use on new shoes and boots shall be sized in B&S (USMC) sizes. Type I heels shall be furnished in sizes 24 through 42, as specified. Type II heels shall be furnished in sizes 27 through 41, as specified. Heels furnished to Government agencies for repair use shall be dealer sized. Type I shall be furnished in sizes 7-8 through 13-14, as specified. Type II shall be furnished in sizes 7-8 through 14-15, as specified (see 6.2).

1.2.2.2 Type III. Type III cuban heels shall be furnished in sizes 1, 3, 5 and 7, as specified; type III junior heels shall be furnished in sizes J-14 through J-17, as specified (see 6.2).

1.2.3 Colors. Rubber heels shall be of the following colors, as specified (see 6.2):

Color No. 1 - Black.
Color No. 2 - Tan.
Color No. 3 - Deleted.
Color No. 4 - Deleted.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing.

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Federal Standards:

FED-STD-123 - Marking for Domestic Shipment (Civil Agencies).
FED-STD-601 - Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Ft. Worth, Denver, San Francisco, Los Angeles and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING:

ARMY NATICK LABORATORIES

8-1-160 - Heels, Rubber; Type II, Style 2 (Full Beveled Breast).

(Miniature copy of Drawing 8-1-160, identified as figure 1, is attached for information purposes only.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

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American Society for Testing and Materials (ASTM) Standards:

D 1630-61 - Abrasion Resistance of Rubber Compounds for Soles and Heels.
D 2240-68 - Indentation Hardness of Rubber and Plastics by Means of a Durometer.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia, PA 19103.)

US Department of Commerce Commercial Standard:

CS35-61 - Hardwood, Plywood.

(Application for copies should be addressed to Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Classes 1, 2, 3, 5, and 7. Classes 1, 2, 3, 5, and 7 shall be made of natural rubber, synthetic rubber, or a mixture of natural and synthetic rubber. The type rubber used, or the proportions of each type rubber if mixtures are employed, shall be such as to meet the requirements specified for each class, as applicable (see 1.2 and 3.2).

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3.1.2 Class 4. Class 4 heels shall be made from a mixture of synthetic rubber and dry cork of 14/24 grade. The mixture shall consist of 8 ± 2 percent by weight of cork, the balance shall be rubber compound.

3.2 Physical characteristics.

3.2.1 Abrasive index before and after aging.

3.2.1.1 Classes 1, 2, 3, and 4. The abrasive index before aging shall be not less than 50 for classes 1, 2, and 3 and not less than 35 for class 4 heels, when tested as specified in 4.3.3. After aging as specified in 4.3.3, the abrasive index of classes 1, 2, 3, and 4 heels shall be not less than 75 percent of the abrasive index determined on the heels before aging.

3.2.1.2 Classes 5 and 7. The abrasive index for classes 5 and 7 shall be not less than 65 before and after aging when tested as specified in 4.3.3.

3.2.1.3 Subclass (a) heels. The abrasive index for subclass (a) heels shall be not less than 175 before and after aging when tested as specified in 4.3.3.

3.2.2 Hardness (before and after aging). The hardness of the heels, (except classes 5 and 7) shall be not less than 60 nor more than 80. The hardness of classes 5 and 7 shall be not less than 50 nor more than 75 when tested as specified in 4.3.3. After aging as specified in 4.3.3, the hardness of classes 1, 2, 3, and 4 heels shall be not more than 10 points higher than the hardness determined on the heels before aging.

3.2.3 Volume swell (classes 2, 4, 5, and 7). Class 4 heels shall have a volume swell not greater than 65 percent, and classes 2, 5, and 7 shall have a volume swell not greater than 50 percent when tested as specified in 4.3.3.

3.2.4 Nonmarking composition (classes 3, 4, 5 and 7). Classes 3, 4, 5 and 7 shall produce only a slight easily erasable mark on white paper when tested as specified in 4.3.3. The heels shall not contain over three parts of carbon black per hundred parts of rubber.

3.3 Thickness. The thickness of the heels shall be in accordance with table I when measured as specified in 4.3.3.

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TABLE I. Thickness of heels

Heel			Thickness (inch)
Type I	-	Back center.	1/2 ± 1/64
		Center breast.	7/16 ± 1/64
Type II	-	Back center.	7/8 min.
		Center breast.	5/8 min.
Type III	-	Back center.	3/8 min.
		Center breast.	5/16 min.

3.4 Design. The molded heels shall be of commercial design conforming to types and styles as specified (see 1.2), except that design and dimensions of type II, style 2 heels shall conform to Drawing 8-1-160, and the dimensions of the type I heels shall be as specified in figure 2.

3.4.1 Nail holes. Each nail hole shall be provided with a metal washer molded in the heel.

3.4.1.1 Type I (half heels). Classes 1 and 3 half heels shall have 8 holes properly located.

3.4.1.2 Type II, classes 1, 2, and 3. Type II, classes 1, 2, and 3 whole heels shall have 13 nail holes. The 5 rear nail metal washers shall be molded in the heel 1/16 inch lower than the 8 forward nail metal washers. All nail holes shall be properly located.

3.4.1.3 Type II, class 4. There shall be no spaces, nail holes, or metal washers in the heel.

3.4.1.4 Type II, classes 5 and 7. There shall be no nail holes or washers in classes 5 and 7.

3.4.1.5 Type II, style 3. Type II, style 3 heels shall have both breast corners truncated at an angle of 45° ± 5° for a distance of 5/16 ± 1/16 inch on the tread surface.

3.4.1.6 Type III (junior and cuban heels). Unless otherwise specified, each junior heel shall have not less than 7 nail holes. Each cuban heel shall have not less than 5 nail holes. Nail holes for junior and cuban heels shall be properly located.

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3.4.2 Heel cores.

3.4.2.1 Type II, class 5. The wood core shall be a 3-ply plywood, black gum, 3/16 inch thickness. The plywood shall conform to the requirements for type I (exterior), good grade (1), of CS 35-61. The bonding agent between the plies shall be treated with a suitable bactericide or fungicide to inhibit degradation. The exposed face shall be coated with a suitable waterproofing substance. The distance of the core from the rear heel edge shall be 5/16 (+ 1/32, -1/16) inch. A rubber overlap of not less than 5/16 inch shall be bonded and encompass the exterior periphery of the heel core. The core or its plies shall neither delaminate nor separate from the heel when tested as specified in 4.3.3.

3.4.2.2 Type II, class 7. The plastic core shall be not more than 5/16 inch and not less than 1/8 inch from the edge of the heel, from breast corner to breast corner, except that the distance from the breast edge may be closer than 1/8 inch. The depth of the core shall be 1/4 $\pm$ 1/32 inch. The hardness shall be 65 $\pm$ 5 Shore D and not less than 55 Shore D after oil immersion. The core shall be firmly bonded to the rubber and shall show no evidence of separating before and after oil immersion, and no evidence of cracking when tested as specified in 4.3.3. The exposed surface of the core shall be stippled.

3.5 Sizes. Sizes shall be in accordance with 1.2.2 and as specified in the contract or order (see 6.2).

3.6 Color. The color shall be color No. 1 or 2 as specified (see 1.2.3 and 6.2).

3.7 Marking for identification.

3.7.1 Trademark and size. The manufacturer's name or trademark shall be legibly marked on the wearing surface of each heel. Both the B&S size and the dealers size shall be legibly marked on the attaching face of each heel.

3.8 Workmanship. The finished heels shall be clean and well made, and shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQLs).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3.2 Inspection of the end item. The end item shall be examined for the defects listed in 4.3.2.1 and 4.3.2.2. The inspection levels and AQLs shall be as specified in 4.3.2.3. The lot size shall be the number of pairs of heels of the same type, class, style, and color submitted for inspection at one time.

4.3.2.1 Visual examination. The heels shall be examined in accordance with the list of defects set forth below. The sample unit shall be one completely fabricated pair of heels.

Examine	Defect
Pairing	Improperly matched, i.e., not same size, or wide variation in color or appearance.
Size	Not as specified.
Color	Not as specified; streaked, mottled.
Design	Any characteristic not in accordance with the specified requirements.

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Examine	Defect
Type, class, and style	Not as specified.
Construction and workmanship	Any area of porosity or sponginess resulting in loss in density of the rubber stock. More than one cavity or air check 1/16 inch or more deep but not exceeding 1/3 the thickness dimension of the heel. Any cavity or air check more than 1/3 the thickness dimension of the heel in depth. Any press cut, tear, check, blister, loss of tread detail, imbedded foreign matter, or otherwise defective. Not cleanly trimmed. Number of nail holes not as specified.
Identification marking and size marking	Missing; incomplete; incorrect; illegible.
Heel cores (applicable to plywood and plastic)	Material other than specified. Any core omitted, damaged, cracked, delaminated, or improperly bonded.
Plywood core	Heel overlap missing or not bonded as specified. Waterproof coating missing or improperly applied.
Plastic core	Any plastic core showing evidence of separation or softening. Core not stippled.

4.3.2.2 Dimensional examination. The heels shall be examined for defects in dimensions in accordance with the list below. The sample unit shall be one heel.

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Defect

Any dimension that does not conform to figure 2 (type I).
Any dimension that does not conform to Drawing 8-1-160, as applicable (type II, style 2).
Any wood core heel dimension that does not comply with the requirements of 3.4.2.1.
Any plastic core heel dimension that does not comply with the requirements of 3.4.2.2.
Any truncated dimension and angle not as specified in 3.4.1.5 (type II, style 3).

4.3.2.3 Inspection levels and AQLs. The inspection levels and the AQLs expressed in defects per hundred units shall be as follows:

	Inspection level	AQL
For examination in 4.3.2.1	II	4.0
For examination in 4.3.2.2	S-2	4.0

4.3.3 Testing of the end item. The methods of testing contained in FED-STD-601, whenever applicable, and as listed in table III shall be followed. The physical values specified apply to the average of determinations made on a sample unit. All test reports shall contain the individual values utilized in expressing the final result. The lot size shall be expressed in units of one pair of heels. The sample unit for test shall be twelve heels. The sample size (number of sample units) shall be as shown in table II. The lot shall be unacceptable if one or more units fail to meet any requirement specified. For those characteristics where test results are reported as pass or fail, there shall be no failure of any test specimen to meet the requirement as specified. For requirements applicable to the lot average, there shall be no failure to meet the requirement as specified.

TABLE II

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. End item tests

Characteristic	Requirement paragraph	Test method	Requirements applicable to: Individ. unit	Requirements applicable to: Lot aver.	Number determinations per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Composition	3.1.1 and 3.1.2	1/	-	-	-	-	-
Abrasive index before aging	3.2.1	ASTM D 1630-61	-	X	3	-	Unit
Abrasive index after aging	3.2.1	7221 2/ and ASTM D 1630-61	-	X	3	-	Unit
Hardness before aging	3.2.2	3021	-	X	5	-	Scale reading
Hardness after aging	3.2.2	3021 and 7221 2/	-	X	5	-	Scale reading
Volume swell (classes 2, 4, 5, and 7)	3.2.3	6211 3/	X	-	3	-	Percent
Nonmarking composition (classes 3, 4, 5, and 7):							
Marking test	3.2.4	4.4.1	X	-	1	X	-
Carbon black content	3.2.4	1/	-	-	-	-	-

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TABLE III. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method	Requirements applicable to: Individ. unit	Requirements applicable to: Lot aver.	Number determinations per unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest
Thickness	3.3	4/	X	-	3	-	1/64 inch
Plywood cores							
Material	3.4.2.1	1/	-	-	-	-	-
Bond test	3.4.2.1	4.4.2	X	-	1	X	-
Plastic cores:							
Hardness, Shore D durometer:							
- before oil immersion	3.4.2.2	ASTM D 2240-68	-	X	5	-	Scale reading
- after oil immersion	3.4.2.2	4.4.3.2	-	X	5	-	Scale reading
Cracking	3.4.2.2	4.4.3.1	X	-	1	X	-
Bond test	3.4.2.2	4.4.3.2	X	-	1	X	-

1/ The supplier shall furnish a certificate of compliance for each shipment or lot citing conformance to the applicable requirement.

2/ Aging shall be conducted at 212°F for 24 hours.

3/ Specimens shall be cut from heel, shall be 1 by 1 inch, and shall be buffed to a thickness of 0.075 ± 0.010 inch. The immersion medium shall be medium No. 6 fuel of Method 6001 of FED-STD-601.

4/ Cut heel in half lengthwise; use Method 2021 of FED-STD-601 to measure thickness. Alternatively, a ratchet type machinist micrometer may be used without cutting heel.

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4.3.4 Examination of preparation for delivery requirements. An examination shall be made to determine that the packaging, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per 100 units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing or not as specified. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, improper taping, loose strapping, or inadequate stapling. Bulged or distorted container.
Content	Number per interior package or container is more or less than specified.

4.4 Test methods.

4.4.1 Marking test. The edge of the heel shall be drawn across a sheet of white bond paper using moderate pressure. If only a light mark is made which is easily erased by rubbing with the fingers, the heel shall be classified as nonmarking.

4.4.2 Wood core bond test (fuel). The heel with the treated and waterproof coated wood core shall be immersed in medium No. 6 fuel of Method 6001 of FED-STD-601 for four hours at standard conditions. The heel shall then be removed from the fluid and wiped. The overlap shall be lifted or otherwise examined and shall show no evidence of separation, delamination, or lack of bonding. In addition, a portion of the core section of the heel shall be cut out and the wood plies examined for evidence of separation, delamination or lack of bonding.

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4.4.3 Plastic core tests.

4.4.3.1 Cracking, low temperature. The heel containing the plastic core shall be conditioned as specified in Method 5111 of FED-STD-601 to an equilibrium temperature of 0°F ± 1.8°F. The specimen shall be removed after 3 hours. Immediately thereafter, 5 heeling nails shall be immediately hammered by hand into the exposed core to a depth greater than 5/16 inch. The nails shall be placed more than 1/2 inch apart along the rear center and sides and the core shall show no evidence of cracking.

4.4.3.2 Bond and hardness. The stippled surface of the plastic core shall be sanded smooth and the heel shall be immersed in medium No. 6 fuel of Method 6001 of FED-STD-601 for 4 hours at standard conditions. The heel shall be removed from the fluid, wiped, and examined. The core shall show no evidence of separation, delamination, or lack of bonding. The core shall then be tested for hardness on its smoothened surface in accordance with ASTM Method D 2240-68, Shore D durometer.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Types I and II. Packaging not required.

5.1.1.2 Type III. One hundred and forty-four pairs of heels of one description only shall be packaged in a fiberboard interior box conforming to type CF (variety SW), class domestic, grade 175 of PPP-B-636. The approximate inside dimensions of each interior box shall be as specified in table IV. The heels, in pairs, shall be arranged on end 12 in length, 4 in width and 3 in depth within the box. Each box shall be securely closed with 2 inch minimum width gummed paper tape conforming to type III, grade A of PPP-T-45.

TABLE IV. Interior box dimensions for type III heels

Heels	Sizes	Approximate inside dimensions 1/ (inches)
Cuban	1	9-1/2 by 7 by 6
	3	9-1/2 by 8 by 7
	5, 7	9-1/2 by 9 by 7-1/2
Junior	J-14 through J-17	9-1/2 by 11-1/2 by 9-1/4

1/ Furnished as a guide only.

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5.1.2 Level B (Civil agencies). Heels shall be packaged as specified in 5.1.1.

5.1.3 Level C. Heels shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Types I and II. One hundred and forty-four pairs of type I heels or 72 pairs of type II heels of one description only shall be packed in a fiberboard shipping container as hereinafter specified. The approximate inside dimensions of each shipping container shall be as specified in table V. The heels, in pairs, shall be arranged on end within the shipping container as specified in 5.1.1.2, except that type II heels shall be arranged 6 pairs in length.

TABLE V. Shipping container dimensions for types I and II heels 1/

Heels	Sizes	Approximate inside dimensions 1/ (inches)
Type I - half heels	7-8, 9-10	13-1/4 by 11-1/4 by 9-1/4
	10-11, 11-12	13-1/4 by 12-1/2 by 10-1/4
	12-13, 13-14	13-1/4 by 13-3/4 by 11-1/2
Type II - whole heels	7-8, 9-10	12 by 11-3/4 by 9-3/4
	10-11, 11-12	12 by 12-3/4 by 10-1/4
	12-13, 13-14	12 by 13-1/4 by 10-1/2
	14-15	12 by 13-3/4 by 10-3/4

1/ Furnished as a guide only.

5.2.1.2 Type III. Two hundred and eighty-eight pairs of heels of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container as hereinafter specified.

5.2.1.3 Shipping container. Each fiberboard shipping container shall conform to style FOL, grade V2s of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

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5.2.1.4 Mixed sizes. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one description only may be packed within the same container.

5.2.2 Level B.

5.2.2.1 Types I and II. One hundred and forty-four pairs of type I heels or 72 pairs of type II heels of one description only shall be packed in a fiberboard shipping container as hereinafter specified. The approximate inside dimensions of each shipping container shall be as specified in table V. The heels, in pairs, shall be arranged on end within the shipping container as specified in 5.1.1.2, except that type II heels shall be arranged 6 pairs in length.

5.2.2.2 Type III. Two hundred and eighty-eight pairs of heels of one description only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container as hereinafter specified.

5.2.2.3 Shipping container. Each shipping container shall conform to style OSC (minimum 2 inch overlap), type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II as specified in the appendix of the container specification.

5.2.2.3.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.2.4 Mixed sizes. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one description only may be packed within the same container.

5.2.3 Level C. Heels, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

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5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2.1 Labels, mixed sizes. Each shipping container packed with mixed sizes of one description only shall have securely attached to the end and side, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity of pairs and sizes contained therein.

6. NOTES

6.1 Intended use. The heels are for manufacture or repair of boots and shoes.

6.1.1 Type I heels are intended for use on men's dress shoes.

6.1.2 Type II heels are intended for use on field type of men's footwear.

6.1.3 Classes 3, 4, 5, and 7 will not leave a permament mark when scuffed over a clean surface.

6.1.4 Classes 2, 4, 5, and 7 will withstand deterioration when in contact with oil.

6.1.5 Class 4 containing cork is for use where slip-resistance is required.

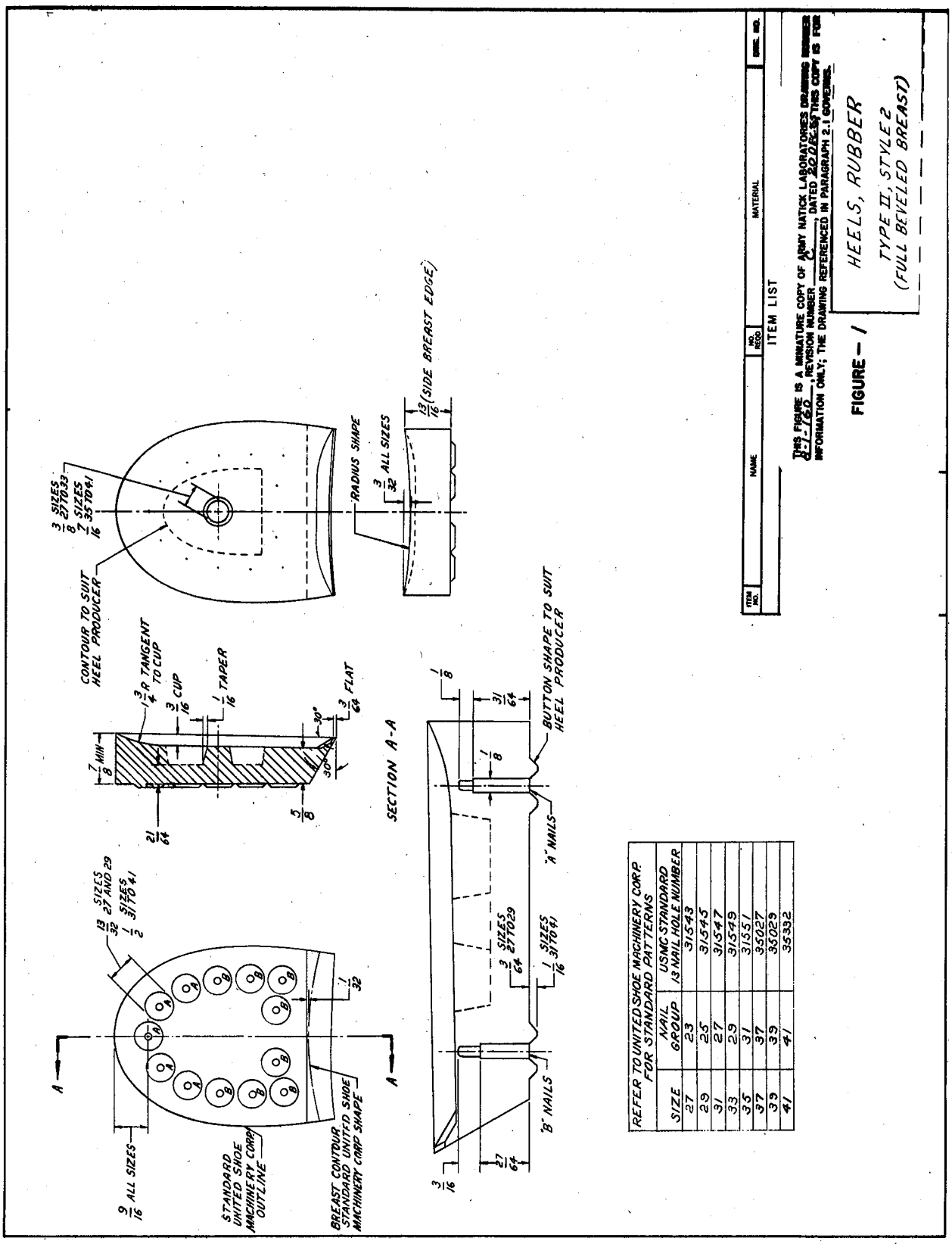
6.1.6 Type I, class 3, subclass (a) heels are high abrasive commercial heels expected to endure the life of the footwear.

6.1.7 Type III heels are intended for use as components for women's semidress shoes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, size, and color of heel required (see 1.2).
(c) When subclass (a) heels are required (see 1.2.1.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B shipments (see 5.2.2.3.1).

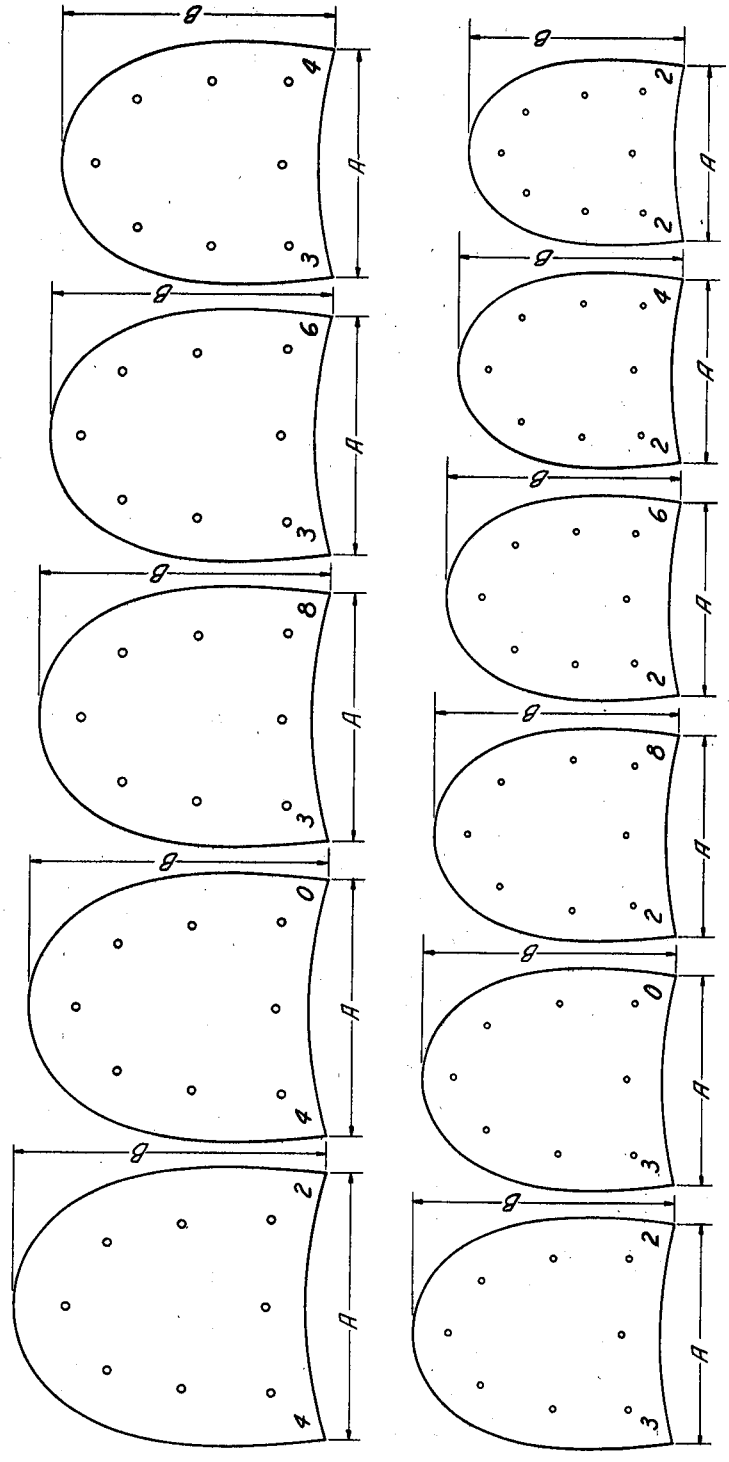
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REFER TO UNITED SHOE MACHINERY CORP. FOR STANDARD PATTERNS

SIZE	NAIL GROUP	USMC STANDARD 13 NAIL HOLE NUMBER
27	23	31543
29	25	31545
31	27	31547
33	29	31549
35	31	31551
37	37	35027
39	39	35029
41	41	35332

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SIZE	A	B
22	2 3/8"	2 1/8"
24	2 1/2"	3"
26	2 5/8"	3 1/8"
28	2 3/4"	3 1/4"
30	2 7/8"	3 3/8"
32	2 7/8"	3 1/2"
34	3 1/8"	3 5/8"
36	3 1/8"	3 3/4"
38	3 5/8"	3 7/8"
40	3 7/8"	4"
42	3 7/8"	4 1/8"

FIGURE – 2 STANDARD DIMENSIONS MEN'S HALF HEELS

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6.3 Correlation between B & S (USMC) sizes and dealer sizes is as follows:

B & S (USMC)	Dealer	
	Type I	Type II
24	7-8	-
25	-	7-8
26	9-10	-
27	-	9-10
28	10-11	-
29	-	10-11
30	11-11	-
31	-	11-12
32	11-12	-
33	-	12-13
34	12-13	-
35	-	13-14
36	13-14	-
37	-	14-15
38	14-15	-
39	-	15-16
40	15-16	-
41	-	16-17
42	16-17	-

Custodians:

Army - GL
Navy - SA
Air Force - 82

Review activity:

Army - MD

User activity:

Navy - CG, MC

Preparing activity:

Army - GL

CIVIL AGENCIES COORDINATING ACTIVITY:

DC - DCG
GSA - FSS
JUS - FPI

Project No. 8335-0050

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DOCUMENT IDENTIFIER AND TITLE

ZZ-H-141D, HEELS, RUBBER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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